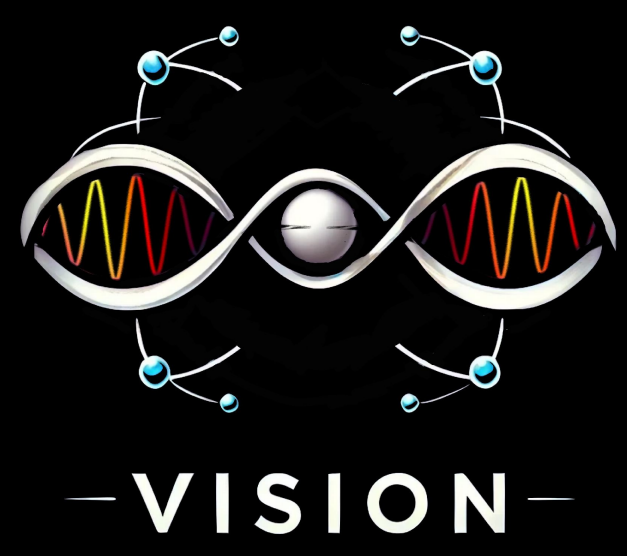




COSMOS—CLUSTER #7-Quantum Materials Robotic Arm Automation for Laser Experiments



Amey Amle^{1,2}, Tanvi Dhoot^{1,3}, Arihant Saxena^{1,4}, Miriam Pena¹, Hui Cai¹, Michael Scheibner¹
University of California Merced¹, Menlo School², The Harker School³, Academy of the Canyons Middle College High School⁴

Introduction

Nitrogen Vacancy (NV) centers give nanodiamonds a quantum spin that is stable at room temperature, making nanodiamonds valuable in modern innovation and technology.¹ This project aims to show how optical tweezers can be used to move and identify these diamonds. This project repurposes a manually operated robotic arm as a “optical tweezer” capable of lifting and holding individual nanodiamonds, roughly 100 nm in diameter. Additionally, this project aims to sort nanodiamonds into two categories: those containing an NV center and those without.

Intended objectives

- Trap a single ~100 nm nanodiamond in a laser focus carried by a robotic arm
- Replace a fixed optical-table with a mobile trap repositionable anywhere in the arm's working radius
- Identify NV centers through photoluminescence collected back through the same objective²
- Sort nanodiamonds into two categories: those containing an NV center and those without³
- Show that a simpler setup can do work that traditionally requires more complex instruments
- Be able to use the arm as a pick and place mechanism for the NV diamonds



Fig. 1: Diagram of Arm with mounted objective

Research Overview + Methods

Optics: 532 nm beam, pointing downward. The first objective started at a 0.95 NA swapped to .65 NA. Neither were able to lift the 100 nm NV diamond³. All $k_B T$ values are calculated at 7.5 mW unless otherwise noted.

Platform: Hiwonder Maxarm, pre-assembled with Python interpreter. The suction apparatus was removed, and a 3D print rigidly connects the object and arm, enclosed in a semi-open laser safe box, repurposed from a CNC

Control: Laptop commands sent to ESP via USB-C cable, able to calibrate torque, speed, and step count, originally tried game controller, but did not get to work

Limits: Servos are accurate to the tenth of a degree, but mechanical limitations, like backlash, do not allow the arm to be very accurate

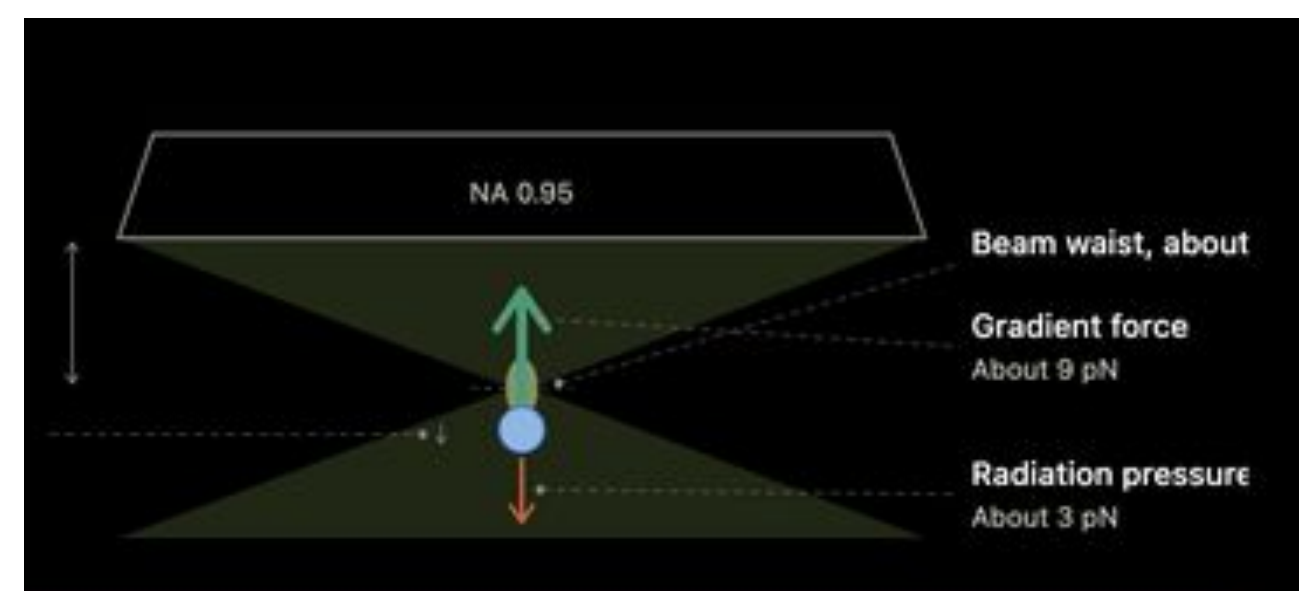


Fig. 2: Force diagram of diamond calculated at 7.5 mW

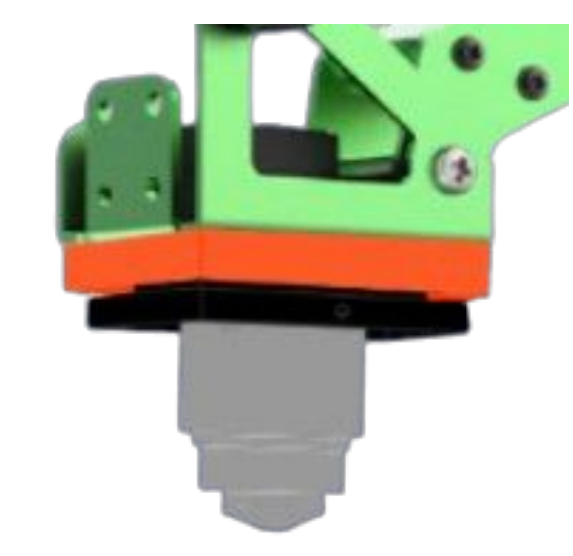


Fig. 3: Diagram of objective

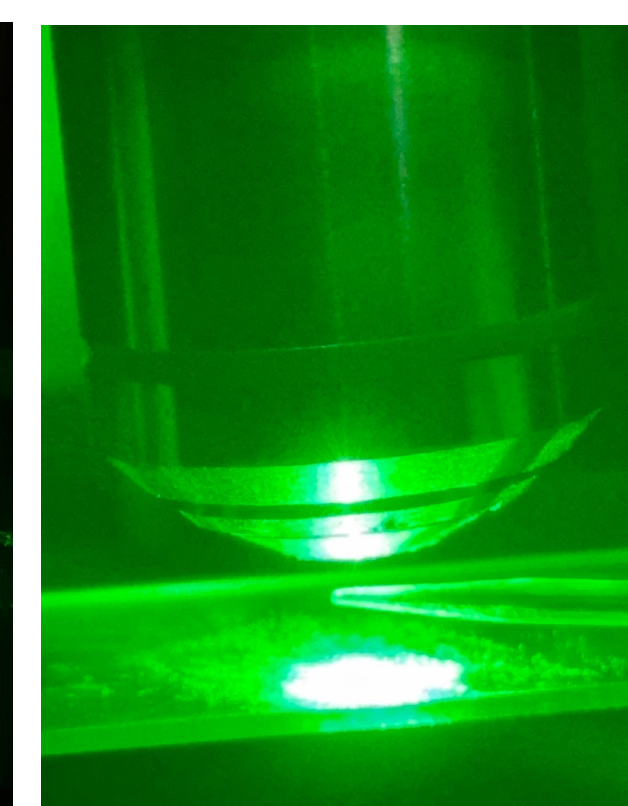
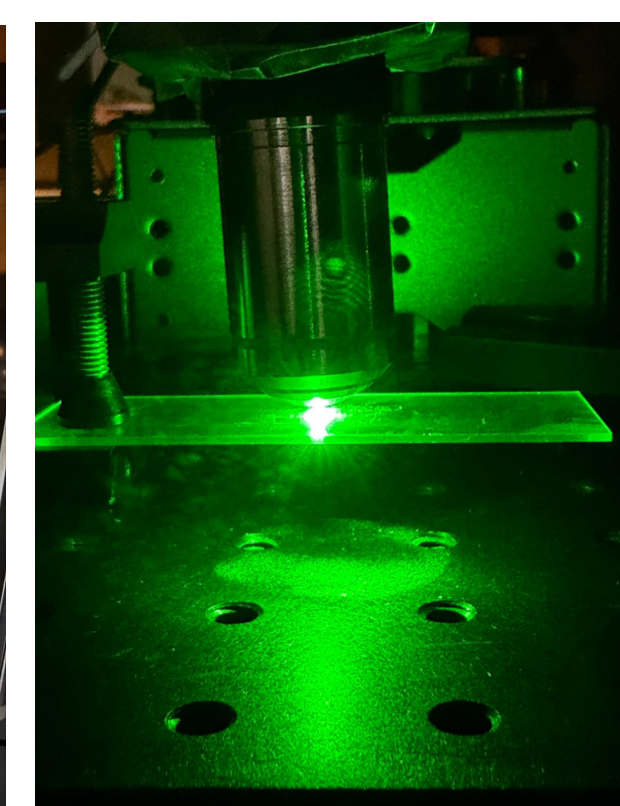
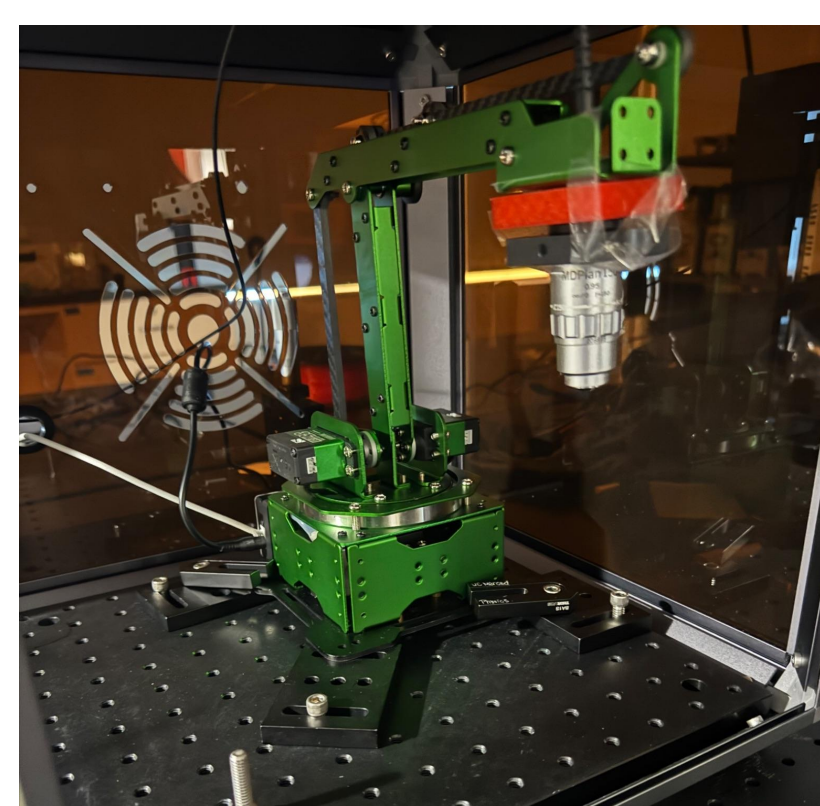


Fig. 4:
Robotic
Arm setup
in enclosure

Fig. 5,6:
Levitiation
attempts

Results

Objective 1 — Transfer arm control and mount objective

- Programming the arm's ESP32 controller directly through HiWonder's Python interpreter, using a Pygame interface
- Original suction assembly and its servo were removed, replaced with a 3D printed mount

Objective 2 — Deliver optical power to the sample

- The initial laser pointer was inadequate in power and beam quality and was replaced with a Excel laser, delivering 7.50-30 mW at 532 nm to the sample¹
- The axial escape barrier is 2.4 $k_B T$ with the 0.65 NA objective. The original 0.95 NA objective would have given 28.2 $k_B T$
- The 0.95 — 0.65 change made to gain working distance cost 92% of axial barrier.
- Use of a multi-mode fiber optic cable led to a dispersing of light; “loose” focus point

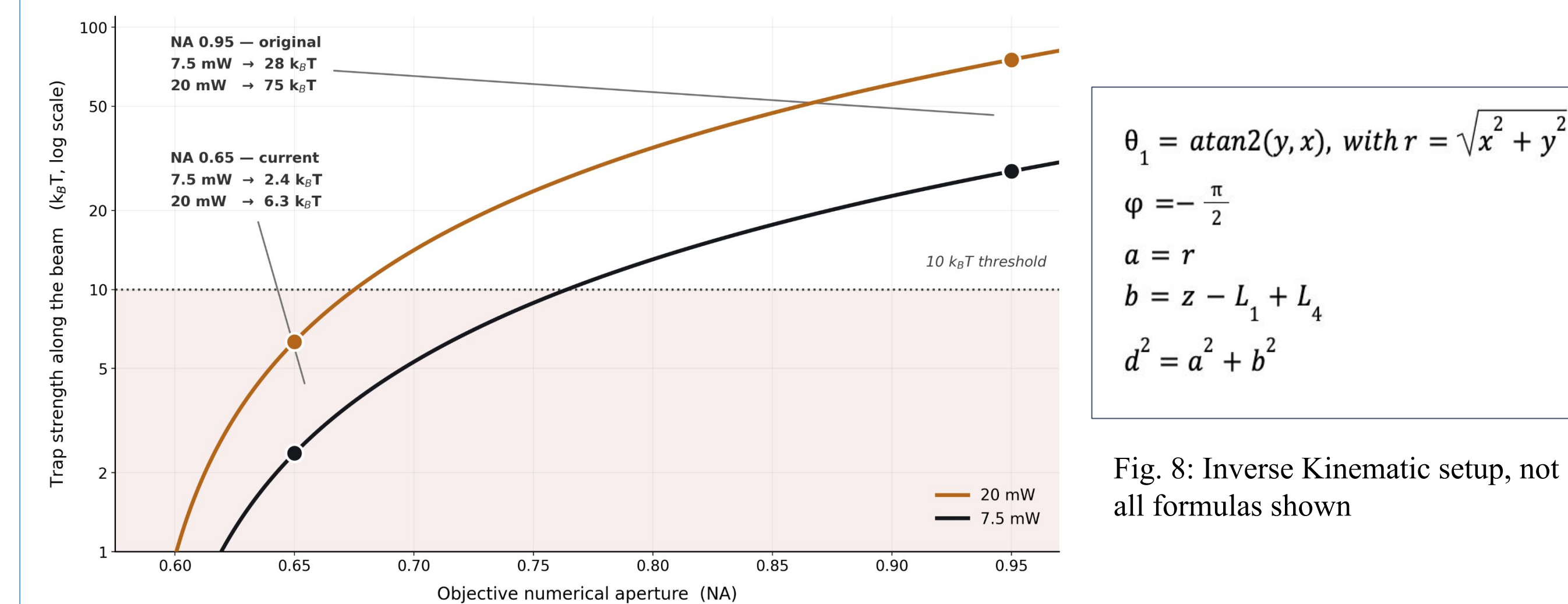


Fig. 7: Comparing Aperture to trap depth at 7.5 mW and 20mW on a logarithmic scale (Ashkin et. al. 1986)

Fig. 8: Inverse Kinematic setup, not all formulas shown

Discussion/Conclusion

- Traded a 0.95 NA objective for 0.65 NA increased working distance from 1.2 mm to 2.4 mm
- NA reduction lowered axial trap depth below the ~10 $k_B T$ threshold needed for stable trapping
- As a result, a nanodiamond could not be fully levitated off the slide in any of the three methods attempted
- In the current setup, the upward gradient force is not strong enough to counteract both the scattering force and gravity, which push the particle downward
- Results are consistent with the low calculated trap depth
- Increasing NA or laser power would fix trap depth, but beam geometry itself may need to shift to trapping from below, so scattering force works with the gradient force instead of against

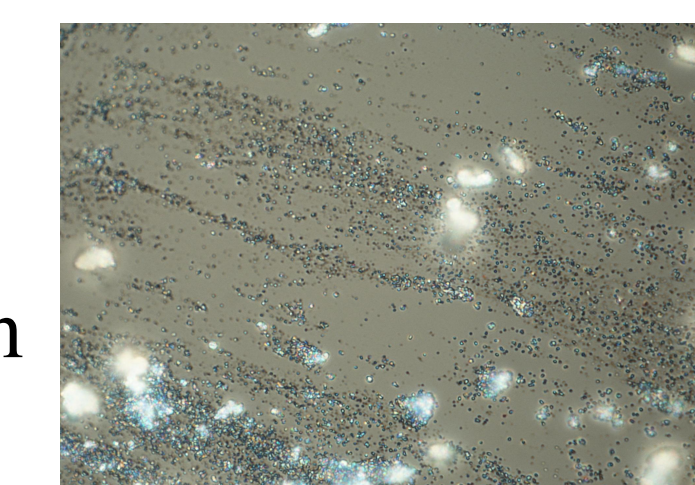


Fig. 9: Nanodiamonds under 50x

$$U_{axial} = \int_{z_{min}}^{z_{barrier}} [F_{grad}(z) - F_{scat}(z)] dz$$

Fig. 10: Modified work integral



Robotic Arm Code Github Link

Future Work

Acquiring a stable trap requires raising the axial barrier above the ~10 $k_B T$ threshold. There are two ways to achieve this: restoring the 0.95 NA objective, or increasing laser power to 32 mW at the current 0.65 NA. The objective route is the more likely path forward, as a 32 mW beam presents an eye hazard in the current semi-open setup. Using the 0.95 NA objective yields a small 1.2 mm working distance constraint but gives 28 $k_B T$, enough to resist thermal escape, leaving only the remaining challenge to mechanical positioning within that clearance.

Milestone 1: Levitate a single nanodiamond at 0.95 NA using a new single-mode cable, which lends to a more concentrated focus

Milestone 2: Move the levitated nanodiamonds with the robotic arm through manual control, showing our precise control over the arm.

Milestone 3: Collect NV fluorescence from the trapped particle through the same objective using a PL spectrometer²

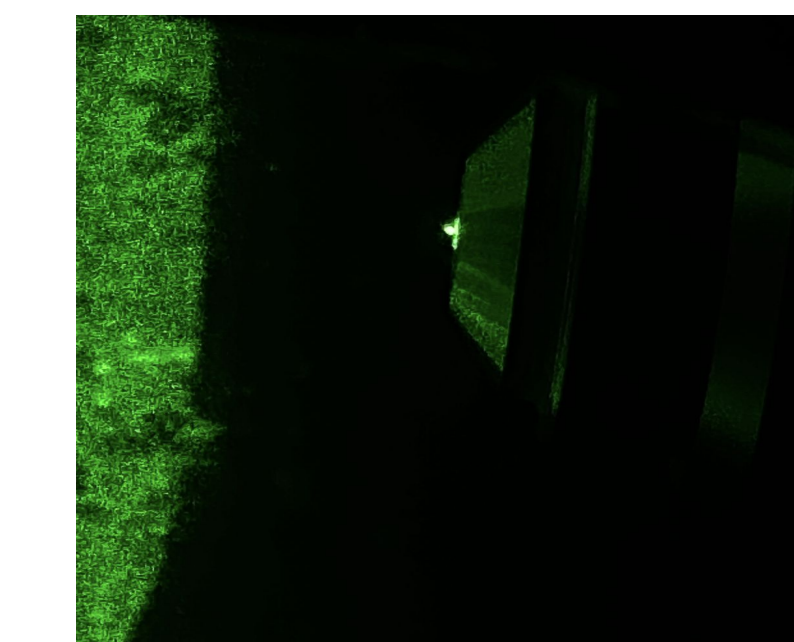


Fig. 11: Beam Waist holding a nanodiamond using a free space coupling

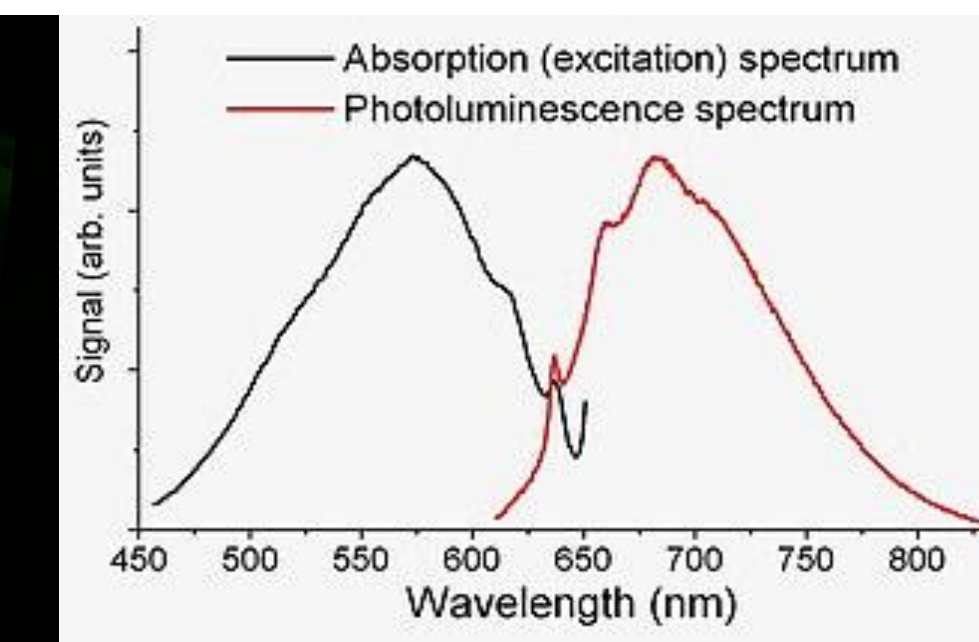


Fig. 12: Expected spectra format (Patel et al., 2025)

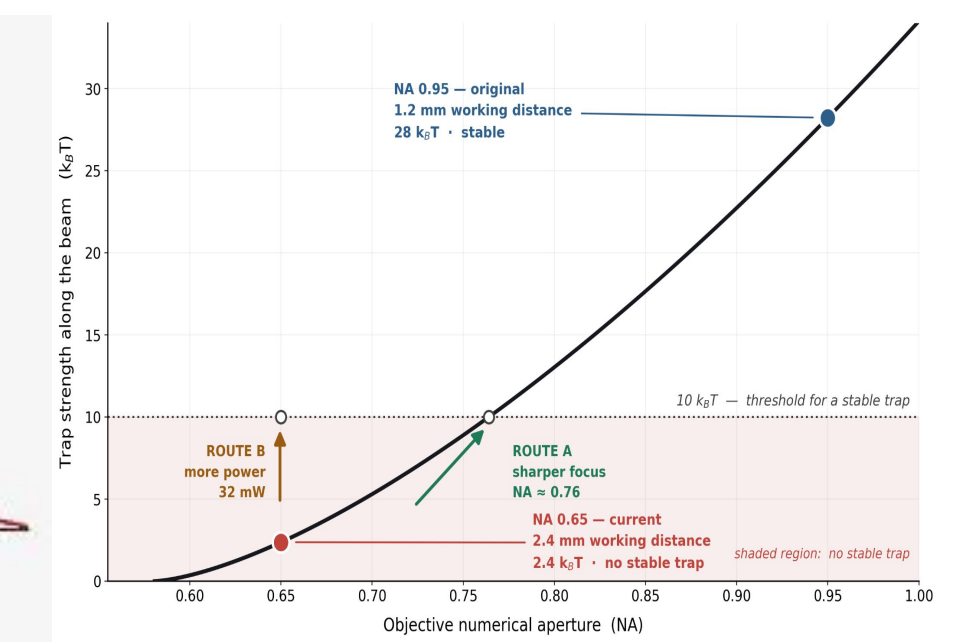


Fig. 13: Graph comparing NA to $k_B T$ at 7.5 mW

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E-Mail: vision-prem@ucmerced.edu
Mailing Address:
VISION-PREM
C/O Prof. Michael Scheibner
Bldg.-Rm.: BSP-045
5200 N. Lake Rd. Merced, CA 95343

